Evaluating Unsupervised Discovery of Persona-Relevant Directions with Matryoshka Sparse Autoencoders

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Abstract

Can sparse autoencoders identify persona-relevant directions without a prespecified target trait? I investigate this question using a Matryoshka BatchTopK sparse autoencoder trained on 100M assistant-position tokens from `Qwen2.5-7B-Instruct`. The autoencoder has approximately 235M parameters and 32,768 features at decoder layer 19, with held-out fraction of variance explained 0.717 and mean $L_0 = 40$. Candidate selection comprises an assistant-specificity arm, a firing-density comparison arm, and a prompt-informed comparison arm. Across 24 scenarios, 60 of 128 features satisfy a diagnostic criterion combining residual separability and cross-scenario consistency; two of three supervised persona-vector controls also satisfy it. A blinded language-model judge produces consistent speaker-related descriptions for 10 signed directions, representing 9 distinct features. All 10 also satisfy the screening criterion. Two originate from selection without persona prompts and eight from prompt-informed selection. However, inspection of the two prompt-free candidates reveals substantial confounding by output degradation. One produces only punctuation in all 24 scenarios, with the judge attributing the named property to the baseline response; the other frequently fails the coherence proxy. The study therefore identifies behaviour-changing candidates but does not yet establish coherent persona discovery without target-trait specification. It provides an empirical evaluation of the discovery question raised by Chen et al. [1] and identifies limitations of the screening and naming criteria. Prompt reachability was omitted from the final run; the ease of prompt elicitation remains a question for future work.

1 Introduction

Persona vectors represent behavioural traits as directions in a language model's activation space. The extraction method of Chen et al. [1] begins with a specified trait and uses contrasting prompted responses to estimate its direction. Their limitations section identifies dependence on advance trait specification, an adequate natural-language description, and successful prompt elicitation. It proposes sparse autoencoders (SAEs) as a complementary route to discovering persona-relevant directions.

Experiment 3C examines discovery before trait naming. I train an SAE to reconstruct residual-stream activations, select candidate features, intervene along their decoder directions, and obtain descriptions from blinded comparisons of generated responses. The central question is whether this process can identify coherent persona-relevant changes without a target trait serving as the candidate-selection criterion.

Dictionary learning and candidate selection require separate interpretation. The entire SAE dictionary is learned without trait labels. One selection arm, however, uses manually specified persona prompts. Evidence for discovery without persona prompts must therefore come from the other arms. I use *persona-relevant* for effects concerning voice, stance, self-presentation, or relation to the reader. Such a description does not by itself establish a complete persona, a distinct psychological category, or a selective intervention.

2 Background and contribution

BatchTopK SAEs constrain the number of active features across a batch while allowing sparsity to vary between tokens [2]. Matryoshka SAEs train nested dictionary prefixes to reconstruct the same activations, encouraging representations at multiple levels of granularity [3]. I combine these methods to construct the feature dictionary used here.

Appendix M of Chen et al. [1] already examines SAE features selected by alignment with specified persona vectors. The present study evaluates selection without that trait-specific alignment, alongside density-based and prompt-informed comparisons. Random directions and supervised persona vectors provide reference measurements. Blinded descriptions assess whether changes concern the speaker. The contribution is this experimental evaluation and the accompanying analysis of its failure modes, rather than a claim to have established unlabelled persona discovery or priority over all other discovery methods.

3 Methods

3.1 Model, data, and autoencoder

The base model is `Qwen2.5-7B-Instruct`. Activations are collected at the output of decoder layer 19 using zero-based indexing. The SAE contains approximately 235M parameters and nested dictionary widths of 1,024, 4,096, 16,384, and 32,768. BatchTopK uses $k = 40$, specifying average batch sparsity rather than exactly 40 active features at every token.

Training uses 100M assistant-position tokens. The configured mixture draws conversations from LMSYS-Chat-1M, `monology/pile-uncopyrighted`, and an insecure-code dataset with sampling weights 0.70, 0.22, and 0.08, respectively. These are conversation-sampling weights, not measured token fractions. Plain-text samples are placed in assistant turns. Downstream screens reuse the trained checkpoint rather than training a new SAE for each screen.

3.2 Candidate selection

The final screen contains 128 distinct features in three disjoint arms:

- **Assistant-specificity (48 features).** Selection uses assistant-versus-user activation statistics, breadth across conversations, topic-invariance proxies, and dictionary structure, without persona prompts or a trait-specific target description.
- **Density-quantile (32 features).** Features are sampled across firing-density quantiles as a comparison that also does not use persona prompts.
- **Prompt-shift (48 features).** Selection uses activation variation across manually specified persona-style contexts. This arm is prompt-informed and does not independently test discovery without persona prompts.

The assistant-specificity arm depends on researcher-designed ranking heuristics; absence of trait labels does not imply absence of inductive bias.

3.3 Steering and controls

Both signs of each feature are evaluated. Calibration tests doses of 0.25, 0.5, 1.0, and 2.0 relative to its recorded maximum activation. A dose is accepted when at least 75% of calibration scenarios pass the coherence proxy; the largest accepted dose is selected, with a fallback when none qualifies. The final screen compares steered and baseline responses across 24 scenarios. Calibration does not guarantee coherent output on the screening scenarios.

Three supervised control vectors are constructed from differences in mean activations under contrasting prompts: evil versus benevolent, sycophantic versus honest, and hallucinating versus factual. These implement a contrastive comparison, not a complete replication of the original persona-vector evaluation. The screen also includes 20 random directions. Random directions use normalized Gaussian vectors scaled by a nominal dose; feature interventions additionally scale by the feature's maximum activation. Their perturbation norms are therefore not matched, limiting the interpretation of the random comparison.

3.4 Diagnostic screening

Leave-one-scenario-out logistic regression measures separability of steered and baseline responses in mean residual and hashed lexical representations. The criterion G_0 uses only residual-view AUC. For signed direction d, let A_d denote this AUC, C_d the mean pairwise cosine similarity between its scenario-specific response differences, and N_d a reference similarity to differences sampled from other directions. The criterion is

$$G_0(d) = \mathbf{1}\left[A_d > \max_{r \in \mathcal{R}} A_r \quad \text{and} \quad C_d - N_d > 0.1\right],$$

where $\mathcal{R}$ contains the 20 random directions. The consistency reference includes other feature, control, and random directions, rather than random directions alone. A feature or control passes if either sign passes. Lexical AUC is supplementary and is not part of G_0.

The implementation reports G_0 diagnostically. It does not automatically block judge evaluation when candidates or controls fail. Control pass counts are therefore reference results rather than certification of the discovery method.

3.5 Blinded description and naming

Judge evaluation selects the 40 highest-ranked signed directions by residual AUC, with consistency as a tie-breaker, and adds the three random directions with the lowest residual AUC. Selection does not require G_0 passage. Each steered-baseline pair is presented in both orders. The judge, `gpt-5.6-luna`, identifies a distinguishing property and classifies it as speaker-, content-, or format-related, or reports no distinction. Feature identities and intervention conditions are withheld from the comparison.

Descriptions are clustered using sentence embeddings and a keyword-overlap merge step. A direction is named when the largest cluster contains at least half of the non-null descriptions, at least 80% of that cluster agrees on which response expresses the property more, and a strict majority is speaker-related. If q_d is the largest cluster's fraction of non-null descriptions and a_d its polarity agreement, the naming score is $S_d = q_d a_d$. The above-null criterion requires S_d to exceed

the highest score of the three judged random directions by 0.1. This is an empirical threshold, not a calibrated significance test.

The naming rule accepts either polarity. A label may describe a property of the baseline response that is absent after steering. The two orderings of each pair are repeated assessments, not independent scenarios. These distinctions are necessary when interpreting the resulting labels.

3.6 Completed scope

Training, ranking, calibration, screening, and blinded description were completed. Prompt reachability was deliberately omitted from the final run. Existing reach-probe code and earlier pilot artifacts do not constitute a final-run reachability result. I reserve prompt-elicitation difficulty for future work.

4 Results

4.1 Reconstruction quality

Held-out fraction of variance explained is 0.717, with mean $L_0 = 40.0$ (Table 1). Training-batch FVE approaches 0.75 (Figure 1); this is distinct from the held-out estimate. The saved reconstruction-replacement check records next-token argmax agreement of 0.780 over 118 assistant positions, below its configured 0.90 threshold. The check therefore fails. Decoder-direction steering does not itself replace the residual stream with the SAE reconstruction, but the failed check limits fidelity claims.

Measure	Recorded value
Held-out FVE	0.717
Held-out mean L_0	40.0
Replacement-check argmax agreement	0.780
Replacement-check cross-entropy increase	0.566 nats
Replacement-check acceptance threshold	0.90 (not met)

Table 1: Saved post-training diagnostics for the checkpoint used in the final screen.

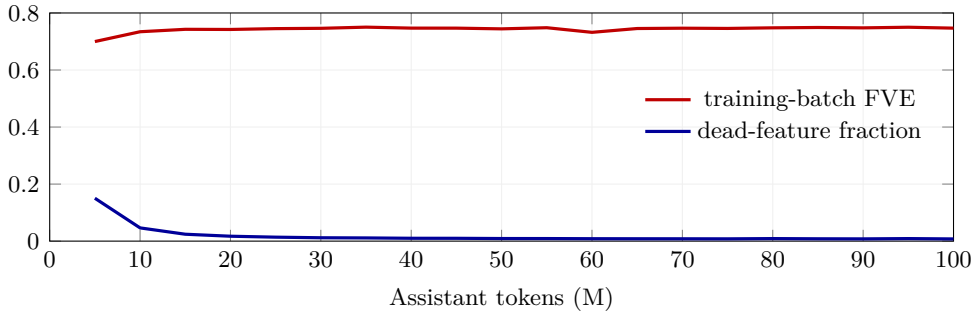


Figure 1: Recorded training diagnostics from run `persona-3c-d77e7c1e`. These checkpoint values are training-batch measurements, not repeated held-out evaluations or independent replications.

4.2 Screening outcomes

An earlier eight-scenario screen records 96/256 features passing G_0 while one of its three controls passes. The final 24-scenario screen records 60/128 features passing in at least one sign, or 79/256 signed directions. The sycophantic and hallucinating controls pass; the evil control does not. This provides positive-control support for two contrasts without establishing sensitivity to all persona-related effects.

The screens differ in candidate count and other settings. Their comparison does not isolate the effect of scenario count, and no formal power analysis is reported. The failed control does not establish that feature counts are conservative or identify why that control failed.

Selection arm	Features	G_0 features	Judged signs	Named signs
Assistant-specificity	48	25	17	2
Density-quantile	32	10	5	0
Prompt-shift	48	25	18	8
Total	128	60	40	10

Table 2: Final screening and naming outcomes. Feature counts group both signs; judge and naming counts refer to signed directions. Unequal, selective judge coverage prevents interpreting naming totals as comparable discovery rates.

4.3 Named directions and output quality

The saved judge output contains 2,063 responses from 2,064 planned comparisons. Each selected feature direction has 48 responses; one random direction has 47. The maximum null naming score is 0.175, yielding an above-null threshold of 0.275. Ten signed feature directions meet the naming criteria and also pass G_0 (Table 3). They correspond to nine features because both signs of feature 2036 are named. Several descriptions also overlap semantically. The count does not establish ten distinct personas.

Feature/sign	Arm	Judge-derived description	Cluster	Agreement
148+	A	engaged and supportive	36/48	1.00
717+	A	composed and confident	26/48	0.96
89−	P	more empathetic and conversational	34/48	0.97
241−	P	formal, academic tone	27/48	1.00
362−	P	empathetic and personally supportive	26/48	1.00
586+	P	enthusiastic and personally affirming	34/48	0.97
709+	P	playful, humorous tone	42/48	0.95
931+	P	formal and self-explanatory	24/48	0.96
2036+	P	formal, academic tone	26/48	0.88
2036−	P	friendly, conversational warmth	33/48	1.00

Table 3: All ten named signed candidates. A: assistant-specificity; P: prompt-shift. Cluster is the largest semantic cluster among non-null descriptions. Agreement is polarity agreement within that cluster, not agreement across all judgements. A label can describe the baseline rather than the steered response.

The two assistant-specificity candidates require particular scrutiny. Feature 148+ produces a single full stop in 23 scenarios and four full stops in the remaining scenario. All 48 judge

comparisons attribute the distinguishing property more strongly to the baseline. Its label therefore describes a contrast with collapsed output, not an induced supportive persona. The coherence proxy nevertheless accepts 23/24 responses because it lacks a minimum substantive response requirement. That high proxy score is not evidence of usable output.

Feature 717+ passes the coherence proxy on only 8/24 scenarios. Its recorded responses exhibit substantial degradation, and the confidence-related label does not establish selective control of confidence. These observations prevent interpreting the two prompt-free named candidates as clean demonstrations of unlabelled persona discovery. The remaining eight named directions originate from prompt-informed selection and require separate quality assessment; they do not resolve the prompt-free claim.

5 Discussion and limitations

Experiment 3C completed the planned discovery and description stages and identified reproducible records of behaviour-changing interventions. However, the central scientific claim requires more than separability and a consistent speaker-related label. Output collapse can satisfy both criteria when the judge consistently describes qualities retained by the baseline. This failure mode is present in the strongest-scoring prompt-free candidate.

The result therefore provides an informative test of SAE-based discovery, but not sufficient evidence that coherent unlabelled personas were discovered. Successful execution of the pipeline and passage of its numerical criteria should be distinguished from validation of the intended construct. Future evaluation should require substantive, coherent responses and assess semantic specificity while explicitly preserving the direction of each judged contrast.

Additional limitations include one model, one layer, one training configuration, and no independent seed replications. Naming relies on one language-model judge and post hoc clustering without independent human validation. The three naming nulls are selected for low residual AUC, whereas evaluated features are selected for high AUC; screening perturbation norms are also unmatched. Finite null maxima and fixed margins do not provide a calibrated false-discovery rate across the searched dictionary. Unequal judge coverage further limits comparisons between selection arms. Leave-one-scenario-out prediction does not turn exploratory method development into an independent confirmatory test.

Prompt reachability is a separate question. Direct activation intervention neither proves nor disproves that prompting can reproduce a behavioural effect. I did not evaluate reachability in the final run. Future work should measure prompt-search attempts, prompt length, and achievable effect size under explicit budgets, matched output quality, and held-out scenarios. This would address ease of elicitation after coherent candidate effects have been established.

6 Conclusion

The trained SAE and completed screening pipeline identify ten named signed candidates across nine features. Two are selected without persona prompts, but inspection reveals output collapse or degradation that confounds their interpretation. The current evidence therefore does not establish coherent unlabelled persona discovery. It extends the empirical investigation proposed by Chen et al. [1] by evaluating a discovery procedure and identifying necessary improvements to its validity checks. The ease of prompt elicitation remains outside the completed study.

Artifacts and authorship

The training run is `persona-3c-d77e7c1e`; the final screen is `persona-3c-ded5477e-a5c3-4372-a776-91a0357293e1`. Code, configurations, and publication sources are available in the AutoLabs repository. The checkpoint is hosted at Hugging Face. Results are derived from the saved `screen_records.json`, `screen_generations.json`, `describe_results.json`, and `post_train_check.json` outputs. Raphael Khalid is the sole researcher and author.

References

[1] R. Chen, A. Arditi, H. Sleight, O. Evans, and J. Lindsey. Persona Vectors: Monitoring and Controlling Character Traits in Language Models. *arXiv:2507.21509*, 2025. `https://arxiv.org/abs/2507.21509`.

[2] B. Bussmann, P. Leask, and N. Nanda. BatchTopK Sparse Autoencoders. *arXiv:2412.06410*, 2024. `https://arxiv.org/abs/2412.06410`.

[3] B. Bussmann, N. Nabeshima, A. Karvonen, and N. Nanda. Learning Multi-Level Features with Matryoshka Sparse Autoencoders. *Proceedings of the 42nd International Conference on Machine Learning*, 267:6077–6101, 2025. `https://proceedings.mlr.press/v267/bussmann25a.html`.